BIOT 6900 — Module 2, Part 3

Alzheimer's Disease Multi-Omics Integration

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Course: BIOT 6900, AI-Powered Biotech Workflows: From Discovery to Clinic

Assignment: Required Module 2 assignment — gene-level integration of Alzheimer's transcriptomic, proteomic, and genomic evidence

Summary of pipeline and results

I integrated three independently sourced Alzheimer's disease (AD) datasets at the gene level:

- **Transcriptomics:** GEO2R differential expression of GSE5281 (Liang et al., 2007; hippocampus, 13 control vs. 10 AD samples), giving log2 fold change and adjusted p-value per probe (54,675 probes, 22,189 gene symbols).
- **Proteomics:** Supplementary Table 2 from Johnson et al. (2022), a TMT mass-spectrometry study of the ROSMAP and Banner brain cohorts, giving AD-vs-control log2 fold change and adjusted p-value per protein (13,541 rows).
- **Genomics:** The GWAS Catalog (Sollis et al., 2023), filtered to Alzheimer's disease associations, giving mapped gene and association p-value per variant (7,410 associations).

These three studies did not use the same patient samples, so it was not possible to compare the results at the individual patient level. As required, I matched all three data layers using gene symbols. The GWAS Catalog sometimes lists multiple genes in one cell, such as TOMM40 and APOE, so I separated them into individual rows. Then, I kept only the most significant result for each gene in each data layer and joined the three tables. Finally, 1,139 genes had usable data in all three layers.

For each gene, I added a concordant column with True or False values. It is True when RNA and protein fold changes show the same direction. Out of 1,139 genes, 560 (49%) showed agreement, which is close to the 50% expected by chance. This also shows that hippocampal RNA and cortical protein data from different patient groups do not always match well. Next, I calculated a multi_evidence_score for each gene. I converted the absolute RNA log2FC, absolute protein log2FC, and −log10 GWAS p-value into ranking percentiles from 0 to 1. Then, I combined the three scores using equal weights, giving each layer one-third of the total score. Finally, I selected the top 15 genes and saved them in a file named targets_ad.csv, were:

Rank	Gene	RNA log2FC	Protein log2FC	GWAS p	Concordant	Score
1	SAMHD1	-3.650	0.054	2e-36	No	0.933
2	PDZD2	-2.290	-0.049	1e-21	Yes	0.898
3	ZMYM2	-2.060	0.053	3e-19	No	0.896
4	APOE	1.170	0.147	< 1e-295*	Yes	0.891
5	ASAP1	-1.480	-0.084	2e-13	Yes	0.871
6	CACNG8	-1.760	-0.096	2e-10	Yes	0.867
7	COL11A1	-1.780	0.365	1e-09	No	0.863
8	MAPT	-1.140	0.180	9e-16	No	0.860
9	APOC1	-1.040	-0.093	< 1e-295*	Yes	0.859
10	RHOJ	-1.500	0.106	6e-11	No	0.856
11	NDUFA10	1.830	-0.071	1e-10	No	0.856

Rank	Gene	RNA log2FC	Protein log2FC	GWAS p	Concordant	Score
12	WNK2	-1.920	-0.041	4e-16	Yes	0.843
13	CLU	-0.918	0.139	1e-47	No	0.841
14	FARP2	-1.820	0.036	2e-42	No	0.840
15	NUMA1	-2.940	0.036	1e-14	No	0.835

*APOE and APOC1 had GWAS p-values stored as exactly 0 (floating-point underflow); these were floored to the smallest non-zero p-value in the data (1e-295) before scoring. Six of the top 15 genes are concordant.

1. Weighting

I kept equal weights, with one-third for transcriptomics, one-third for proteomics, and one-third for genomics. Each layer provides a different type of evidence. GWAS shows inherited genetic risk that may be linked to disease, while transcriptomics and proteomics show changes that happen during disease in affected tissues. Since my goal is to identify genes supported by multiple layers of evidence rather than explain a specific cause-and-effect process, equal weights avoid assuming a mechanism that I have not tested.

Equal weights are meaningful only when all three layers use the same scale. My first attempt used a raw weighted sum. Since −log10 GWAS p-values can reach around 300, while fold changes range from 0 to 4, the ranking was mainly based on GWAS results. To fix this, I converted each layer into rank percentiles between 0 and 1 using multi_evidence_score. This made the three weights genuinely equal.

Rank normalization also has some limitations. It removes differences in magnitude, so APOE's very strong GWAS signal (p below 1e-295) gets the same maximum genomic score as any other top-ranked GWAS gene. As a result, RNA fold change now has a major influence on the top-ranked genes, even though it comes from the smallest dataset, with only 23 samples. For example, SAMHD1 ranks first mainly because it has the largest RNA fold change among the joined genes. One reason to change the weights would be to give more importance to proteomics, which comes from a much larger group and is closer to the actual functional changes. Another option would be to use GWAS as a filter instead of adding it to the score. I kept equal weights to follow the course workflow because there is no single correct weighting method without knowing the specific purpose of the analysis.

2. Top targets

The pipeline identified several well-known AD genes. APOE, the strongest genetic risk factor for late-onset AD, ranks 4th. MAPT (rank 8), which produces tau protein that forms neurofibrillary tangles, is also in the top 15. CLU (rank 13), a well-known GWAS gene involved in amyloid-beta clearance, is also included. APP, the gene for amyloid precursor protein, ranks 19, just outside the top 15, and shows the largest protein increase among the highly ranked genes (log2FC +0.74). Finding these genes is a good sign that the pipeline is identifying real AD-related biology rather than random noise.

APOC1 (rank 9) needs careful interpretation. APOE, APOC1 and TOMM40 are located close together on chromosome 19q13.32 and have strong linkage disequilibrium. Because of this, GWAS studies report signals for all three genes, although APOE, especially the ε4 allele, is thought to be the main cause of the genetic risk. Therefore, APOC1's high genetic score is partly influenced by APOE and should not be considered an independent discovery. TOMM40 shows how much the ranking depends on gene expression data. Even with the same very strong GWAS signal, it ranks 363 because its RNA and protein changes are small.

Among the classic AD genes mentioned in the assignment (APOE, TREM2, BIN1, CLU and PICALM), APOE and CLU are in the top 15. BIN1 (rank 61) and PICALM (rank 221) passed the data joining step but ranked lower because their protein changes are close to zero. TREM2 was completely removed during the joining process. It is present in all three source files, but its row in the proteomics table has no fold change or p-value. Therefore, it was removed when I excluded rows with missing values.

One unexpected gene worth checking further is SAMHD1 (rank 1, discordant). SAMHD1 is not a well-known AD gene. It is an immune regulator that breaks down cellular dNTPs, and loss-of-function mutations cause Aicardi-Goutières syndrome, a serious inflammatory brain disorder linked to type I interferon signaling. In my data, it has the largest RNA change among all joined genes (log2FC −3.65 in the AD hippocampus) and a genome-wide significant GWAS association (p = 2e-36; Willett et al., 2025). Since growing evidence shows that microglial cells and innate immune pathways are important in AD, SAMHD1 could be a promising candidate. However, its protein change is small and moves in the opposite direction. Its high rank also depends heavily on the RNA results from a small dataset. Therefore, independent studies are needed to confirm the finding before considering it a real therapeutic target.

3. Reading a discordant gene

I selected CLU (clusterin, rank 13). Its RNA log2FC is −0.918, meaning its level is lower in AD, while its protein log2FC is +0.139, meaning its level is higher in AD. The protein change is highly significant (adjusted p = 2.8e-10). CLU also has a strong GWAS signal (p = 1e-47). So, CLU shows strong evidence from genetics and RNA data, but its protein level changes in the opposite direction.

Clusterin is a secreted protein that acts as an extracellular chaperone and binds to amyloid-beta. One possible reason for the lower RNA level but higher protein level is that clusterin protein is mainly controlled after transcription. The protein is secreted, remains stable outside cells, and can build up in amyloid plaques. Therefore, the protein level may increase in diseased brain tissue even when RNA production in surviving neurons and glial cells decreases. Also, the two measurements come from different patient groups and brain regions (hippocampus for RNA and prefrontal cortex for protein). So, some of the difference may be due to differences in the patient groups and tissues, rather than one biological reason.

4. Limitation

This analysis was done at the gene level by matching gene symbols across three cohorts that did not share any patients: RNA data from GSE5281 (hippocampus), protein data from the ROSMAP and Banner TMT cohorts (dorsolateral prefrontal cortex), and genetic data from several combined GWAS studies. Since there were no common sample IDs, I could not compare each patient's RNA data with their own protein or genetic data. I can only say that a gene shows changes at the RNA level in one group and at the protein level in another group, which is a much weaker conclusion.

This is different from the matched CPTAC breast cancer case in Part 1, where RNA, protein and mutation data came from the same 122 tumours. This allowed me to calculate the actual RNA–protein correlation for each gene across patients. Here, concordance only compares the average directions of changes from different groups that do not share patients. My results should therefore be understood as showing that these genes have supporting evidence for AD across independent studies, rather than claiming how a gene behaves within a particular patient's brain.

Another limitation is that the score considers only the size of fold changes and does not include statistical significance. For example, APOE has an RNA change with an adjusted p-value of 0.125, which is not statistically significant on its own, but it still contributes to the score. Only about 35% of the 1,139 combined genes have significant RNA changes (adjusted p-value below 0.05). A stricter approach would

be to filter genes based on significance first or use signed −log10 p-values to calculate the score for each data layer.

Data quality notes

Some p-values in the GWAS Catalog were recorded as exactly 0 because of floating-point calculation limits. Before calculating −log10, I replaced these zero values with the smallest non-zero p-value in the dataset. This avoided infinite scores, but the genomic scores for these genes became conservative estimates. Separating GWAS entries that contained multiple genes increased the number of genes in the three-way join from 731 to 1,139. This step was necessary to calculate scores correctly for genes such as APOE.

References

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